



Auto Water Dispenser For Pets

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A pet is loved like a baby in a family and needs lots of attention and care. But sometimes due to our busy schedule we forget to take care of the pet's daily needs. This device will help in providing clean and fresh drinking water to the pet automatically when needed.

Circuit diagram of the auto water dispenser is shown in Fig. 1. It is built around 15V AC step-down transformer X1, bridge rectifier BR1, 12V voltage regulator IC1, Arduino Uno Board1, ultrasonic sensor HC-SR04, water level sensor SEN18, two 5mm LEDs (LED1 and LED2), 2N2222 transistor T1, 12V relay RL1, and 12V DC fluid pump P1. Their functions are described below.

Arduino Uno. Arduino Uno board is connected to water-level sensor and ultrasonic sensor, which detect water level and pet's presence near the water bowl, respectively. When ultrasonic sensor detects presence of the pet nearby, it provides a signal to Arduino.

The Arduino's program compares the received signal with the programmed value and gives signal to its digital output pin 13 to operate (energise) relay RL1, if the programmed conditions are fulfilled. Otherwise, the relay remains de-energised.

12V power supply. It powers Arduino Uno and water pump and helps control the relay. It comprises 230V to 15V, 2A step-down transformer X1, full-bridge rectifier BR1, capacitors C1 and C2, 12V voltage regulator IC 7812, 2.2k resistor R1, and LED1 as indicator. The 15V AC from transformer is converted to DC by the rectifier and is filtered by the two capacitors. IC 7812 regulates the voltage to 12V DC.

Water level sensor SEN18. This sensor has ten exposed conducting tracks, of which five alternate strips are power tracks and the other five are sensing tracks. When the sensor is submerged in water (vertically), bridges are created between the five pairs

of power and sensing tracks due to water's conductivity. Thus, the sensor works like a variable resistor whose value at the sensing pin changes with the change in water level. The value is high when the water level is high and low when water level is low. The sensor can be powered with 3.3V or 5V on-board DC supply.

Relay. This 12V SPDT electromagnetic switching device provides complete electrical isolation between the controlling and output circuits. It is controlled/triggered by a low DC voltage, but its contacts can handle a high current flow with total electrical isolation from the delicate low-power electronic circuit. The relay switches the water pump on and off on receipt of electrical pulse.

Ultrasonic sensor HC-SR04. This device having ultrasonic transmitter and receiver detects presence of an object 2cm to 400cm away. Its ultrasonic transmitter produces 40kHz signal

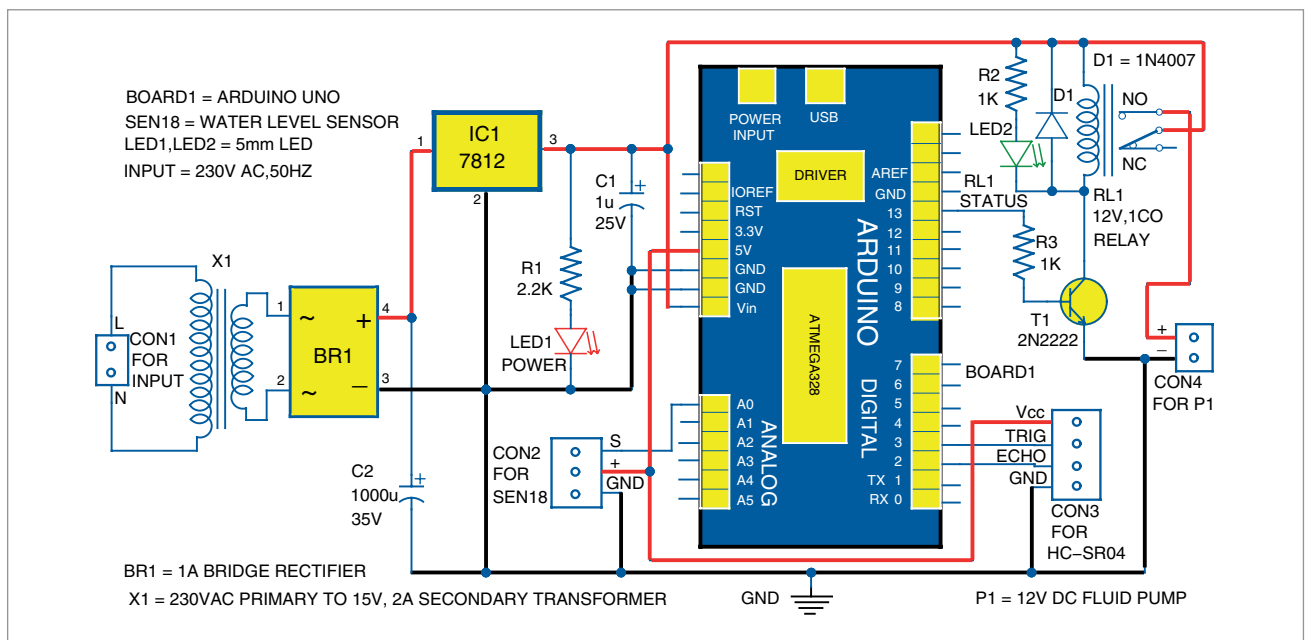


Fig. 1: Circuit diagram of auto water dispenser